Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.22	20.07	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.01	-0.7%
power (W), mean	629	641	+1.9%
power (W), peak	654	682	+4.4%
energy (Wh)	212	214	+1.1%
energy, parked workers still on at idle power (Wh)	212	214	+1.1%
CPU used (cores), mean	0.892	1.20	+34.0%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.037	0.050	+34.0%
energy per core-hour (Wh)	705	536	-23.9%
node-hours per core-hour	6.73	5.02	-25.4%
pending pods, pod-minutes	0.267	0.000	-100.0%
pending pods, peak	1.00	0.000	-100.0%
HPA replicas, mean	8.96	9.28	+3.5%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	1.07	+36.2%
requests timed	1,040	1,100	+5.8%
time over the response line (% of samples)	1.25	0.000	-100.0%
response time (ms), mean	156	95.14	-39.2%
response time (ms), median	129	81.20	-37.2%
response time (ms), 95th percentile	309	154	-50.0%
response time (ms), 99th percentile	519	185	-64.4%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.4190	5.4632	-26.4%	[-3.6271, -0.2966]	GAIN
kwh_per_core_hour	0.7740	0.5803	-25.0%	[-0.3590, -0.0314]	GAIN
utilisation	0.0367	0.0507	+38.2%	[+0.0014, +0.0280]	GAIN
pending_pod_minutes_per_hour	0.7559	0.0000	-100.0%	[-2.2677, +0.0000]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	2.2677	3.3729	+48.7%	[-3.7795, +6.7192]	STEADY, no significant difference

What Omni-Compass did

- decisions: 120 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 4
- power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.327 0.333 0.34 0.346 0.353 0.359 0.365 0.371 0.376 0.382 0.387 0.392 0.397 0.402 0.407 0.412 0.416 0.421 0.425 0.43 0.434 0.438 0.442 0.446 0.45 0.454 0.458 0.462 0.465 0.469 0.472 0.476 0.479 0.482 0.485 0.488 0.491 0.494 0.497 0.499 0.502 0.505 0.508 0.51 0.513 0.516 0.518 0.521 0.523 0.525 0.528 0.53 0.532 0.534 0.537 0.539 0.541 0.542 0.544 0.546 0.548 0.55 0.551 0.553 0.554 0.556 0.557 0.558 0.559 0.56 0.561 0.562 0.562 0.563 0.564 0.565 0.566 0.566 0.567 0.568 0.569 0.57 0.57 0.571 0.572 0.573 0.574 0.575 0.575 0.576 0.577 0.578 0.578 0.579 0.58 0.58 0.581 0.582 0.582 0.583 0.584 0.584 0.585 0.585 0.586 0.586 0.587 0.587 0.587 0.587 0.588 0.588 0.588 0.588 0.589 0.589 0.589 0.589 0.59 0.59
- nodes decided per minute: 6 6 6 6 6 6 6 6 6 6 6 5 5 5 5 5 5 4 4 4 4 4 4 4 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	616	621	5.00	5.00	0.000	0.000	0.557	0.680
1	6.00	6.00	616	625	5.00	6.75	0.000	0.000	0.560	0.792
2	6.00	6.00	619	621	5.00	7.00	0.000	0.000	0.605	0.667
3	6.00	6.00	623	633	5.75	7.50	0.250	0.000	0.718	1.01
4	6.00	6.00	633	642	9.25	10.00	0.000	0.000	1.00	1.27
5	6.00	6.00	632	645	10.00	10.00	0.000	0.000	1.01	1.32
6	6.00	6.00	633	647	10.00	10.00	0.000	0.000	0.985	1.37
7	6.00	6.00	649	676	10.00	10.00	0.000	0.000	1.39	2.15
8	6.00	6.00	648	672	10.00	10.00	0.000	0.000	1.39	2.01
9	6.00	6.00	647	668	10.00	10.00	0.000	0.000	1.41	1.91
10	6.00	6.00	642	660	10.00	10.00	0.000	0.000	1.26	1.70
11	6.00	6.00	621	625	10.00	10.00	0.000	0.000	0.674	0.770
12	6.00	6.00	618	622	10.00	10.00	0.000	0.000	0.625	0.725
13	6.00	6.00	626	633	10.00	10.00	0.000	0.000	0.824	0.789
14	6.00	6.00	633	646	10.00	10.00	0.000	0.000	0.989	1.36
15	6.00	6.00	634	645	10.00	10.00	0.000	0.000	1.02	1.32
16	6.00	6.00	630	648	10.00	10.00	0.000	0.000	0.926	1.39
17	6.00	6.00	627	638	10.00	10.00	0.000	0.000	0.838	1.12
18	6.00	6.00	619	624	10.00	10.00	0.000	0.000	0.627	0.689
19	6.00	6.00	618	626	10.00	10.00	0.000	0.000	0.575	0.795

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.